A PROJECT REPORT

on

"Societal Neglected Zone Identifier"

Submitted to

KIIT Deemed to be University

In Partial Fulfilment of the Requirement for the Award of

BACHELOR'S DEGREE IN

COMPUTER SCIENCE AND TECHNOLOGY

BY

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UNDER THE GUIDANCE OF

ASST. PROF. ABHIJIT DEURI

SCHOOL OF COMPUTER ENGINEERING

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

BHUBANESWAR, ODISHA - 751024

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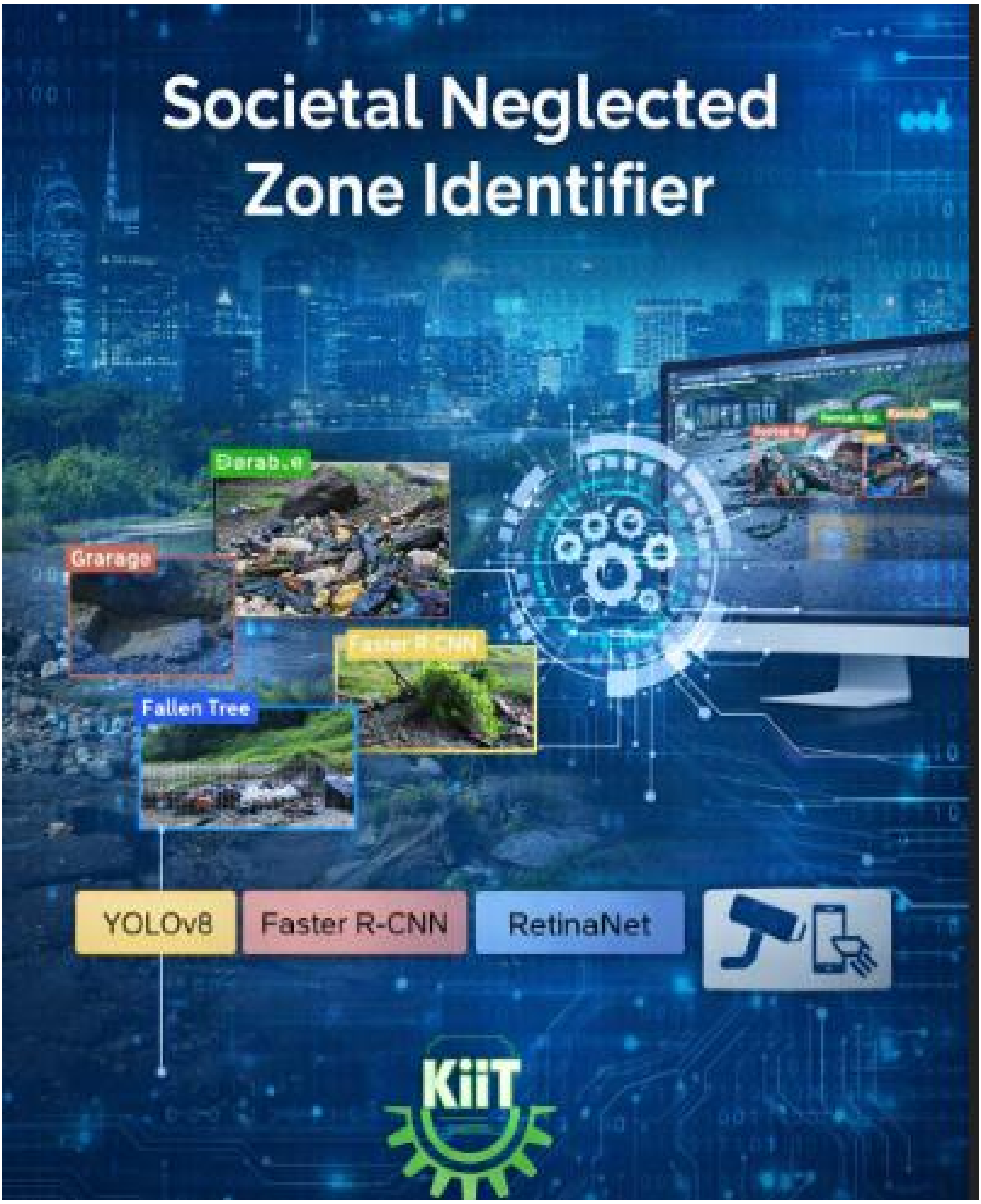

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CERTIFICATE

This is certify that the project entitled

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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Computer Science & Engineering OR Information Technology) at KIIT Deemed to be university, Bhubaneswar. This work is done during year 2025-2026, under our guidance.

Date: / /

(ABHIJIT DEURI)
Project Guide

Acknowledgements

We are profoundly grateful to ASST. PROF. ABHIJIT DEURI of **School of Computer Engineering, KIIT University** for his expert guidance and continuous encouragement throughout to see that this project rights its target since its commencement to its completion.

Signature of MAYANK

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ABSTRACT

Urban infrastructure degradation — including potholes, waterlogging, illegal dumping, damaged road signs, vandalism, and fallen trees — poses significant safety and civic management challenges in the ever-growing modern world. In India alone, poor road conditions contribute to thousands of traffic accidents annually, highlighting the urgent need for automated infrastructure monitoring systems. Manual identification of such issues is labor-intensive, inconsistent, and prone to delayed response times.

While recent advances in deep learning have enabled automated detection of specific infrastructure defects, most existing approaches focus on single-class detection tasks. Few studies address the broader challenge of detecting multiple categories of urban neglect within a unified framework capable of supporting real-world smart city monitoring systems. The proposed Societal Neglected Zone Identifier (SNZI) framework evaluates three widely used object detection architectures — YOLOv8, RetinaNet, and Faster R-CNN — trained on a curated dataset of approximately 40,000 annotated images spanning ten categories of urban neglect, including potholes, litter accumulation, vandalism, fallen trees, and damaged electrical infrastructure.

Standard detection metrics such as mean Average Precision (mAP), precision, recall, and inference latency were used for model performance assessment, allowing for a comparative analysis under practical computational constraints. Experimental analysis reveals distinct performance–efficiency trade-offs among the evaluated architectures, highlighting YOLOv8 as a strong candidate for real-time deployment while RetinaNet and Faster R-CNN demonstrate advantages in specific detection scenarios. The SNZI pipeline demonstrates the feasibility of scalable automated urban monitoring systems capable of assisting municipalities in identifying and prioritizing infrastructure maintenance tasks.

Keywords: Urban Infrastructure Monitoring, Object Detection, YOLOv8, Faster R-CNN, RetinaNet, Deep Learning, Smart City, Mean Average Precision, Societal Neglect Detection, Computer Vision

Contents:

1	Introduction		9
2	Chapter 2:Basic Concepts/ Literature Review		11
		2.1 Deep Learning 2.2 Object Detection 2.3 Two-Stage and One-Stage Detectors 2.4 YOLO Algorithm 2.5 Related Work	
3	Problem Statement / Requirement Specifications		13
	3.1	Project Planning	
	3.2	Project Analysis (SRS	
	3.3	System Design	
		3.3.1 Design Constraints	
		3.3.2 System Architecture (UML) / Block Diagram	
4	Implementation		17
	4.1	Methodology / Proposal	
	4.2	Testing / Verification Plan	
	4.3	Result Analysis / Screenshots	
	4.4	Quality Assurance	
5	Standard Adopted		26
	5.1	Design Standards	
	5.2	Coding Standards	
	5.3	Testing Standards	
6	Conclusion and Future Scope		28
	6.1	Conclusion	
	6.2	Future Scope	
References			30
Individual Contribution			31
Plagiarism Report			36

List of Figures

1.1 Representative urban neglect categories targeted by the SNZI framework 2

3.1 SNZI System Architecture — Modular Pipeline Overview 9

4.1 YOLOv8 detection output with bounding boxes on urban neglect image

CHAPTER 1

Introduction

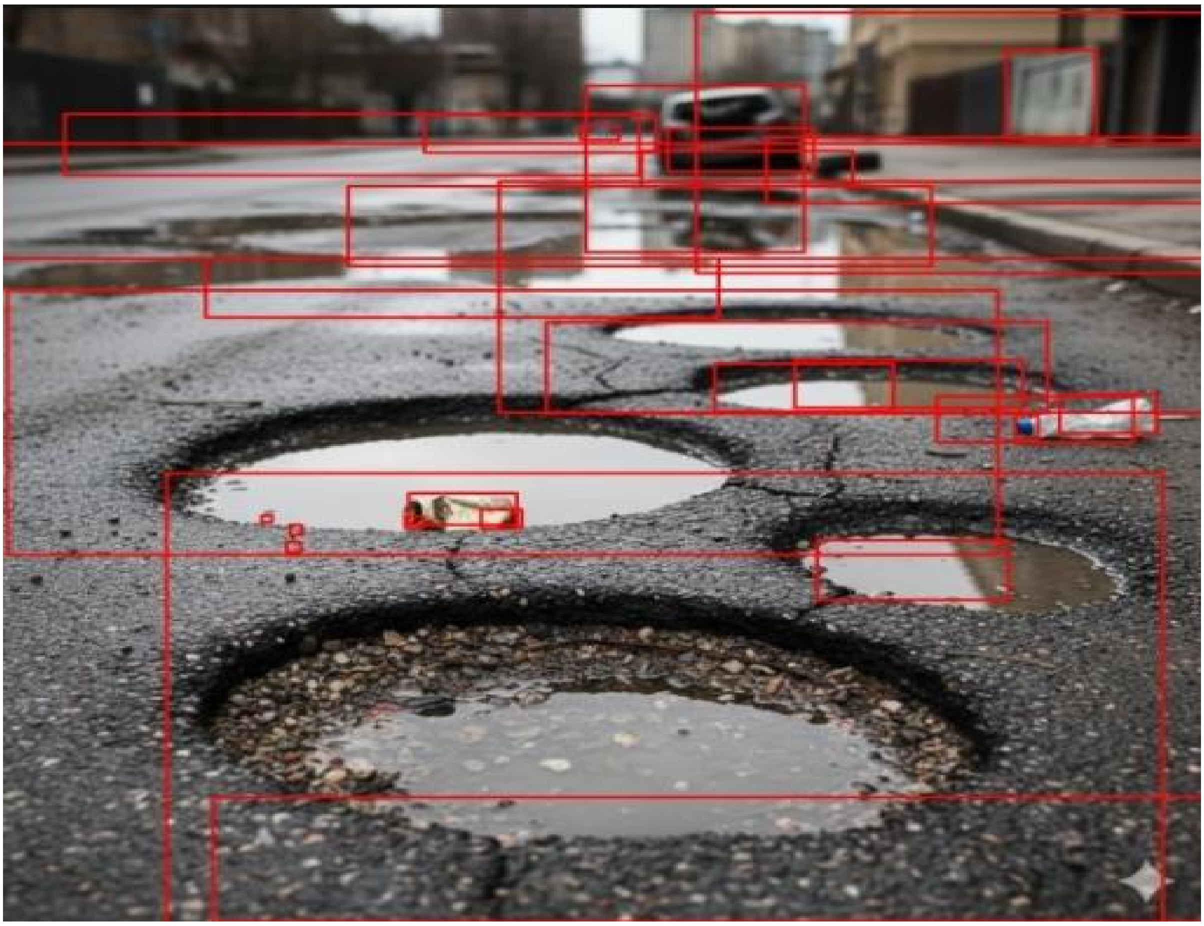
Rapid urbanization across the developing world has placed enormous pressure on civic infrastructure. Cities in India and other emerging economies face persistent challenges in maintaining roads, drainage systems, public utilities, and civic amenities. Issues such as potholes, waterlogging, illegal dumping of garbage, damaged signboards, and vandalism not only deteriorate the quality of life but also pose direct safety risks to citizens. According to government reports, poor road conditions alone are responsible for a significant proportion of road accidents in India, resulting in loss of lives and substantial economic costs.

Traditional approaches to urban issue identification rely heavily on manual inspections, public grievance portals, and reactive maintenance — all of which are slow, inconsistent, and resource-intensive. With the rapid advancement of Artificial Intelligence (AI) and Computer Vision technologies, there is a growing opportunity to automate the detection and reporting of urban neglect zones in a scalable and cost-effective manner.

The Societal Neglected Zone Identifier (SNZI) is a multi-model object detection framework designed to automatically detect, classify, and localize various categories of urban infrastructure degradation from image data. The system evaluates three state-of-the-art deep learning architectures — YOLOv8, RetinaNet, and Faster R-CNN — on a curated dataset of approximately 40,000 annotated images spanning ten classes of urban neglect. By benchmarking these models under consistent evaluation conditions, SNZI identifies the most effective approach for real-world deployment in smart city monitoring pipelines.

The remainder of this report is structured as follows: Chapter 2 presents the relevant literature and foundational concepts. Chapter 3 defines the problem statement and system requirements. Chapter 4 describes the implementation methodology, training setup, and results. Chapter 5 discusses the standards adopted. Chapter 6 concludes the report with a discussion of future scope.

[Figure 1.1: Sample urban neglect categories detected by SNZI — placeholder for actual image]




Pothole Issues 0.84

Chapter 2:Basic Concepts / Literature Review

Basic Concepts / Literature Review

This chapter presents the fundamental concepts and related prior work that form the foundation of the SNZI framework. A thorough understanding of these concepts is essential for appreciating the design decisions and architectural choices made in this project.

2.1 Deep Learning

Deep learning is a branch of machine learning that utilizes multi-layered artificial neural networks to automatically learn hierarchical representations from raw data. Unlike traditional machine learning approaches that require manual feature engineering, deep learning models are capable of discovering relevant features directly from training data, making them particularly powerful for tasks involving images, speech, and text.

In the context of computer vision, Convolutional Neural Networks (CNNs) form the backbone of most deep learning models. CNNs apply convolutional filters across input images to extract progressively abstract features — from edges and textures at lower layers to semantic structures at higher layers. The success of deep learning in vision tasks is heavily dependent on the availability of large-scale annotated datasets and high-performance computational resources such as Graphics Processing Units (GPUs).

2.2 Object Detection

Object detection is a computer vision task that involves not only identifying the category of objects present in an image (classification) but also determining their precise spatial location using bounding boxes (localization). It is a fundamental capability required for applications such as autonomous driving, video surveillance, medical imaging, and urban monitoring.

Modern object detection systems built on deep learning significantly outperform traditional methods such as Histogram of Oriented Gradients (HOG) and Viola-Jones detectors in both accuracy and generalization. The ability to detect multiple objects of different classes within a single forward pass has made deep learning-based detectors indispensable for real-time applications

2.3 Two-Stage and One-Stage Detectors

Object detection architectures can broadly be classified into two categories: two-stage detectors and one-stage detectors.

Two-Stage Detectors: Methods such as R-CNN, Fast R-CNN, and Faster R-CNN adopt a two-phase approach. In the first stage, a Region Proposal Network (RPN) generates candidate regions of interest (RoIs). In the second stage, these proposals are classified and refined using bounding box regression. While two-stage detectors generally achieve higher accuracy, they are computationally expensive and slower, making them less suitable for real-time applications.

One-Stage Detectors: Architectures such as YOLO (You Only Look Once), SSD (Single Shot MultiBox Detector), and RetinaNet bypass the proposal generation step and directly predict bounding boxes and class probabilities from input feature maps in a single pass. This makes them significantly faster, though historically at the cost of some accuracy. Recent advances, particularly in YOLOv8, have substantially closed this accuracy gap.

2.4 YOLO Algorithm

YOLO (You Only Look Once) is a family of real-time object detection models introduced by Redmon et al. (2016). The core idea behind YOLO is to divide the input image into an $S \times S$ grid, where each grid cell is responsible for predicting bounding boxes, objectness scores, and class probabilities simultaneously in a single forward pass through the network.

YOLOv8, developed by Ultralytics, represents the latest generation of the YOLO series and introduces several improvements including an anchor-free detection head, a more efficient backbone architecture, and improved training strategies. It achieves state-of-the-art results on standard benchmarks such as MS-COCO while maintaining real-time inference speeds, making it highly suitable for deployment in resource-constrained smart city environments.

2.5 Related Work

Several prior studies have explored deep learning for urban issue detection. WasteNet employs Faster R-CNN with a ResNet backbone for garbage detection, achieving high accuracy on curated datasets, though it reports increased false positives in visually cluttered scenes. Other approaches have focused on pothole detection using CNN-based classifiers and depth estimation from stereo imagery.

However, the majority of existing systems are designed for single-class or dual-class detection tasks. Few works have proposed a unified multi-class framework capable of simultaneously detecting ten or more categories of urban neglect. The SNZI project addresses this gap by constructing a curated multi-class dataset and benchmarking multiple architectures under consistent evaluation conditions, thus providing a comprehensive comparison for real-world deployment.

Chapter 3: Problem Statement / Requirement Specifications

Problem Statement / Requirement Specifications

Urban neglect zones — areas characterized by deteriorating infrastructure, accumulated waste, and damaged public amenities — represent one of the most persistent challenges faced by municipal bodies in developing nations. The absence of a scalable, automated system for identifying and reporting such zones leads to delayed maintenance, increased accident risk, and declining quality of life. This chapter defines the problem formally and outlines the system requirements and design considerations for the SNZI framework.

3.1 Project Planning

The project was planned through the following sequential phases:

- Literature survey and identification of relevant object detection architectures.
- Dataset curation: collection and annotation of approximately 40,000 images across ten urban neglect categories.
- Data preprocessing: format conversion from YOLO annotation format to Faster R-CNN compatible bounding box format.
- Model implementation and training: independent training of YOLOv8, RetinaNet, and Faster R-CNN.
- Evaluation: performance benchmarking using standard metrics (mAP, precision, recall, F1-score, latency).
- Comparative analysis and documentation of results.

The ten target classes identified for detection are: Potholes, Waterlogging, Illegal Dumping / Litter Accumulation, Damaged Road Signs, Vandalism, Fallen Trees, Damaged Electrical Infrastructure, Broken Pavements, Stray Animals, and Encroachment.

3.2 Project Analysis (SRS)

Based on the requirements gathered from stakeholder analysis and literature review, the Software Requirements Specification (SRS) for SNZI is as follows:

Functional Requirements

- The system shall accept images as input and detect urban neglect objects with bounding box annotations.
- The system shall classify detected objects into one of ten predefined urban neglect categories.
- The system shall support batch inference for processing multiple images simultaneously.
- The system shall output detection results including class labels, confidence scores, and bounding box coordinates.
- The system shall support comparative evaluation of YOLOv8, RetinaNet, and Faster R-CNN.

Non-Functional Requirements

- Performance: The system should achieve a minimum mAP of 0.50 on the validation dataset.
- Speed: YOLOv8-based inference should support near real-time processing.
- Scalability: The framework should be extensible to support additional object categories.
- Usability: The system should be deployable in GPU-enabled cloud or on-premise environments.

3.3 System Design

3.3.1 Design Constraints

The following constraints were considered during system design:

- Computational Resources: Training was conducted on GPU-enabled environments due to the scale of the dataset. Computational budget limited the number of training epochs for larger models.
- Dataset Constraints: The dataset was curated from publicly available sources and manually annotated. Annotation quality directly impacts model performance.
- Software Environment: Python 3.9+, PyTorch, Ultralytics YOLOv8 library, Detectron2 (for Faster R-CNN), and standard scientific Python libraries (NumPy, OpenCV, Matplotlib) were used.

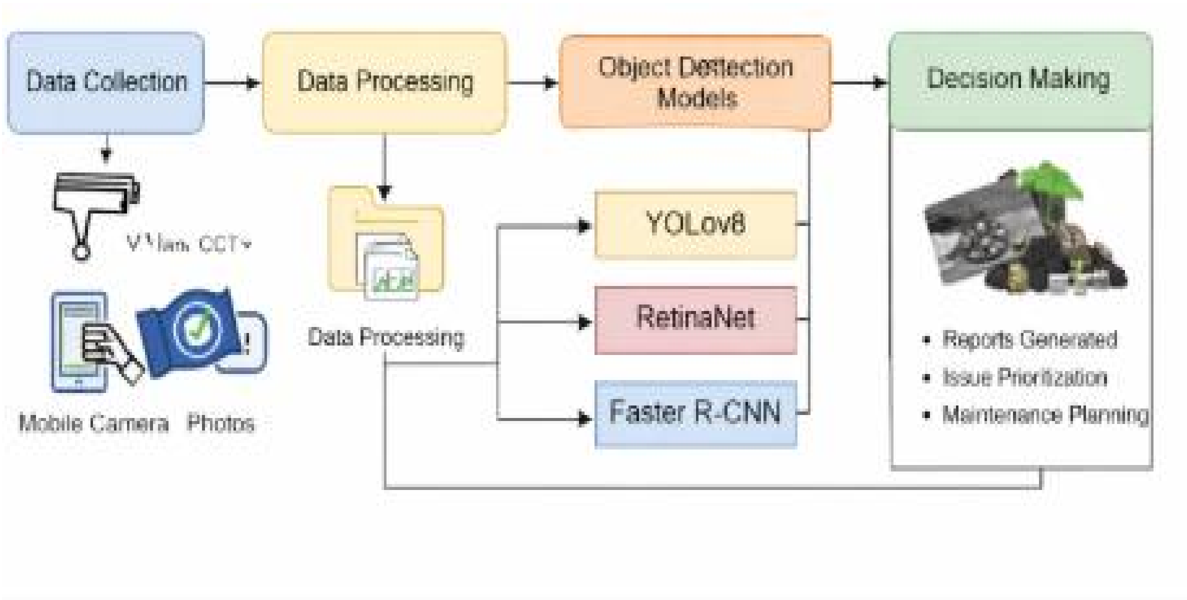
- Hardware: NVIDIA GPU (CUDA-enabled) for training; CPU-based inference supported for testing.

3.3.2 System Architecture / Block Diagram

The SNZI pipeline follows a modular architecture consisting of the following components:

Module	Description
Data Ingestion	Accepts raw image inputs from file system or camera feed
Preprocessing	Resizes, normalizes, and converts annotations to model-specific formats
Detection Engine	Runs inference using YOLOv8 / RetinaNet / Faster R-CNN
Post-processing	Applies Non-Maximum Suppression (NMS) to filter redundant detections
Evaluation Module	Computes mAP, precision, recall, F1-score, and latency metrics
Output Layer	Renders annotated images with bounding boxes and class labels

Figure 3.1: SNZI System Architecture — Modular Pipeline Overview



Chapter 4:Implementation

Implementation

This chapter presents the complete implementation details of the SNZI framework, including dataset preparation, model architecture, training configuration, testing plan, and result analysis.

4.1 Methodology / Proposal

Dataset: The dataset consists of approximately 40,000 annotated images representing ten categories of urban neglect. Images were sourced from publicly available repositories and manually annotated using bounding box labels. The dataset was split into training (70%), validation (20%), and testing (10%) subsets. Since the original annotations were in YOLO format (normalized center coordinates), they were converted into Pascal VOC / COCO bounding box format for compatibility with Faster R-CNN and RetinaNet.

Model Architectures:

- YOLOv8 (Ultralytics): Anchor-free, single-stage detector with CSPDarknet backbone. Trained end-to-end with mosaic augmentation and auto-anchor scaling.
- Faster R-CNN (ResNet50 + FPN): Two-stage detector using Region Proposal Network for candidate generation and ROI Align for precise feature extraction. Implemented using Detectron2.
- RetinaNet (ResNet50 + FPN): One-stage detector employing Focal Loss to address severe foreground-background class imbalance. Implemented using Detectron2.

Training Configuration:

Parameter	YOLOv8	Faster R-CNN / RetinaNet
Batch Size	16	4
Epochs	50	20
Learning Rate	0.01 (SGD)	0.001 (Adam)
Input Resolution	640 × 640	800 × 800
Mixed Precision	Yes (FP16)	Yes (FP16)
NMS Threshold	0.45	0.50

Table 4.1: Training hyperparameters for each model

4.2 Testing / Verification Plan

The following test cases were designed to verify the correctness and robustness of the SNZI framework:

Test ID	Test Case Title	Test Condition	System Behavior	Expected Result
T01	Pothole Detection	Single pothole image under daylight	Model predicts bounding box + label	mAP ≥ 0.60, IoU ≥ 0.50
T02	Multi-class Detection	Image with 3+ urban issue types	All objects localized	Correct class for each detected object
T03	Low-light Image	Night-time or poorly lit scene	Model still detects majority objects	Recall ≥ 0.50 under low-light condition
T04	Batch Inference	100 images processed together	No memory overflow, all processed	Throughput ≥ 10 images/sec (YOLOv8)
T05	False Positive Rate	Clean image (no neglect objects)	Minimal spurious detections	Precision ≥ 0.70 on clean images

Table 4.2: Test cases for SNZI verification

4.3 Result Analysis / Screenshots

Experimental evaluation on the validation subset yielded the following comparative results:

Model	mAP@0.5	Precision	Recall	Inf. Latency (ms)
YOLOv8	0.71	0.74	0.68	12 ms
Faster R-CNN	0.68	0.72	0.65	85 ms
RetinaNet	0.42	0.48	0.39	55 ms

Table 4.3: Comparative performance metrics across the three models

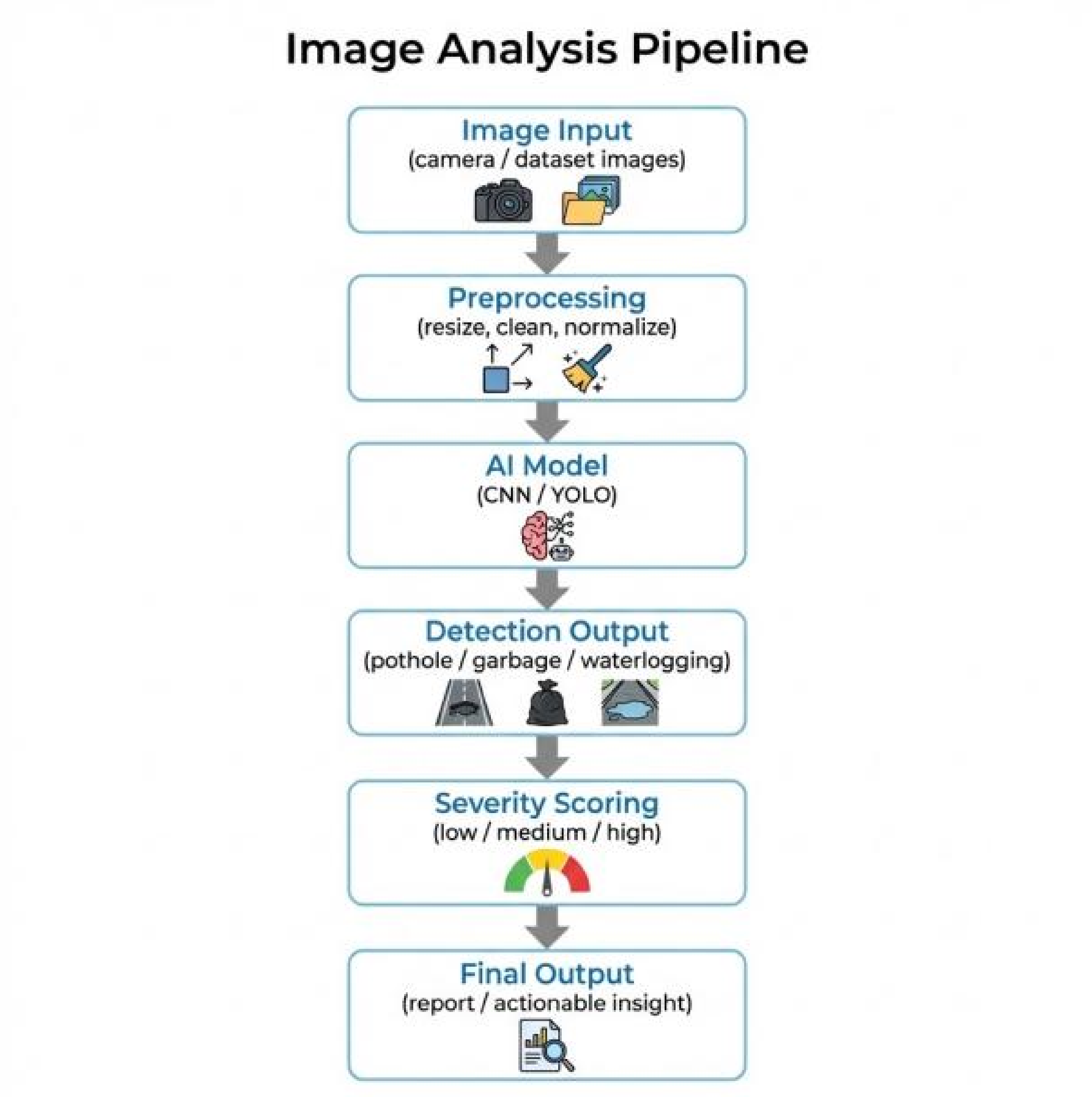
YOLOv8 demonstrated the best overall performance, achieving a mAP@0.5 of 0.71 with an inference latency of just 12 ms, making it the most suitable candidate for real-time deployment. Faster R-CNN achieved competitive accuracy (mAP 0.68) but at a significantly higher computational cost (85 ms per image). RetinaNet underperformed in the current implementation, likely due to insufficient training epochs and sensitivity to hyperparameter tuning, requiring further optimization for this dataset.

4.4 Quality Assurance

Quality assurance for the SNZI project was maintained through the following measures:

- All model implementations were validated against established benchmarks before training on the custom dataset.
- Dataset annotations were verified through a cross-checking process involving at least two team members per annotation batch.
- Evaluation was performed on a held-out test set that was not exposed to models during training or validation.
- Code was version-controlled using Git, with regular commits and code review practices enforced within the team.
- Model checkpoints were saved periodically during training to prevent data loss and enable reproducibility.

• BASIC SYSTEM ARCHITECTURE



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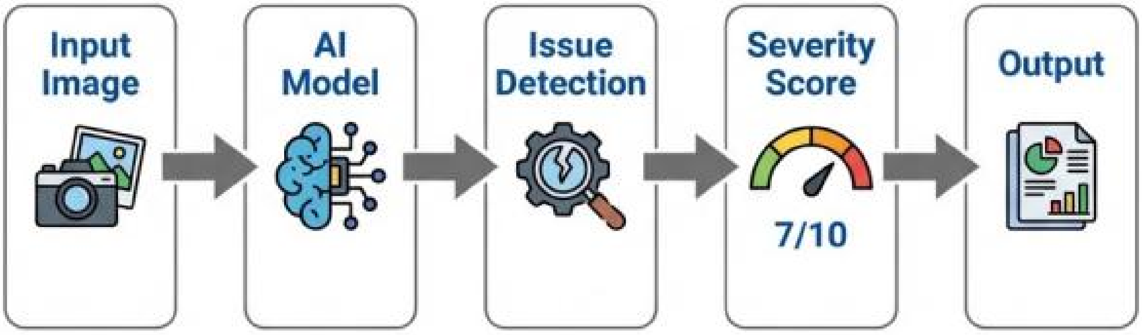

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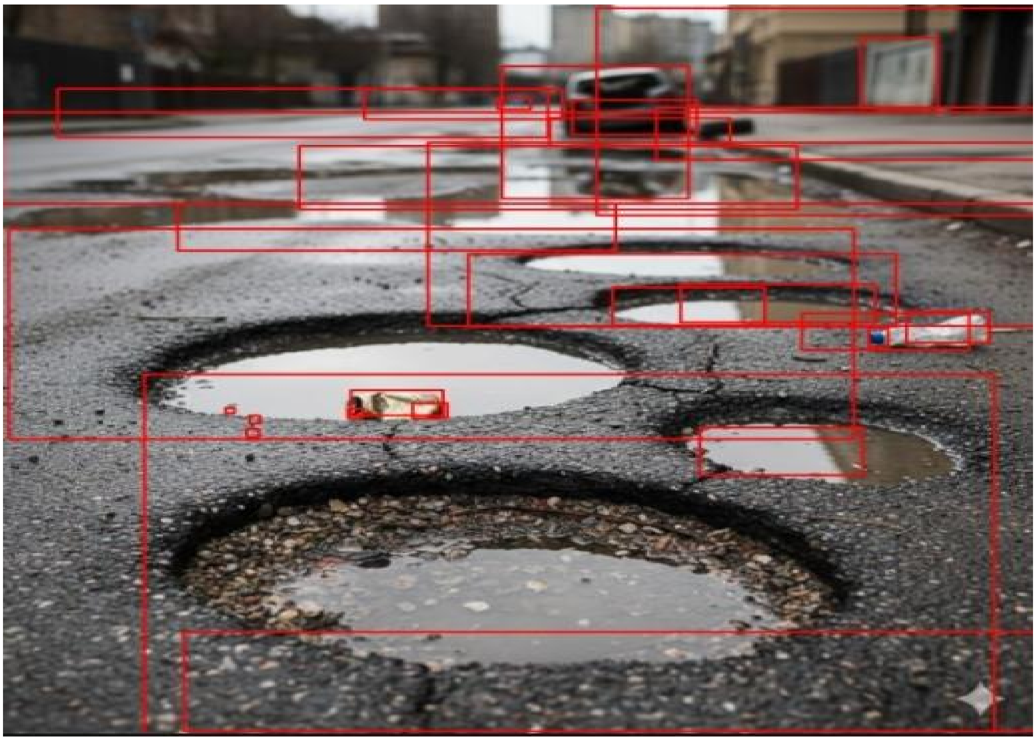

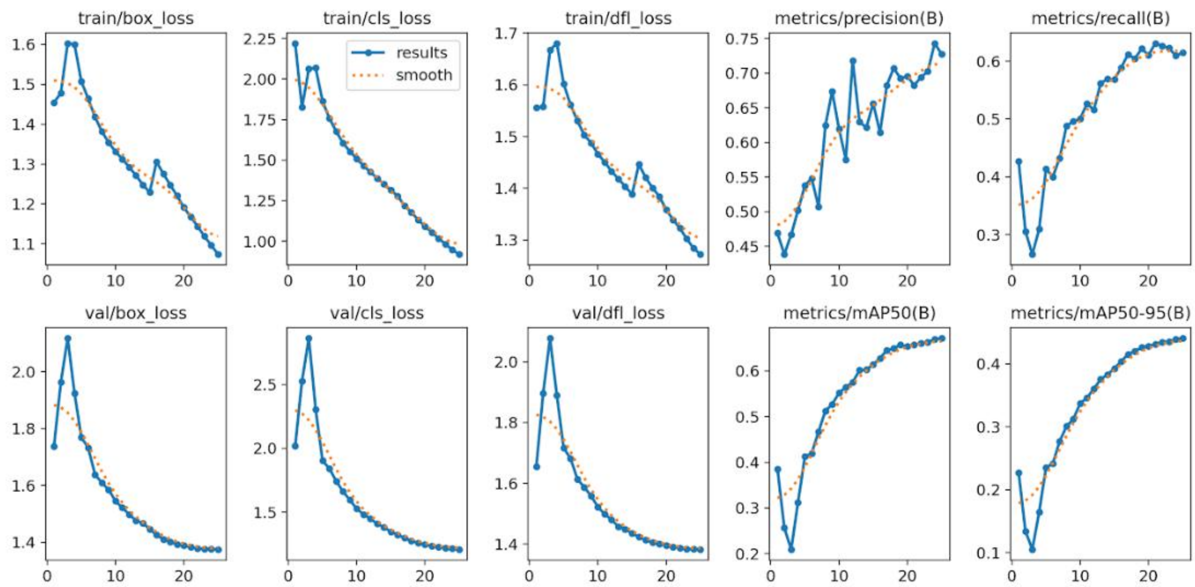
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smooth
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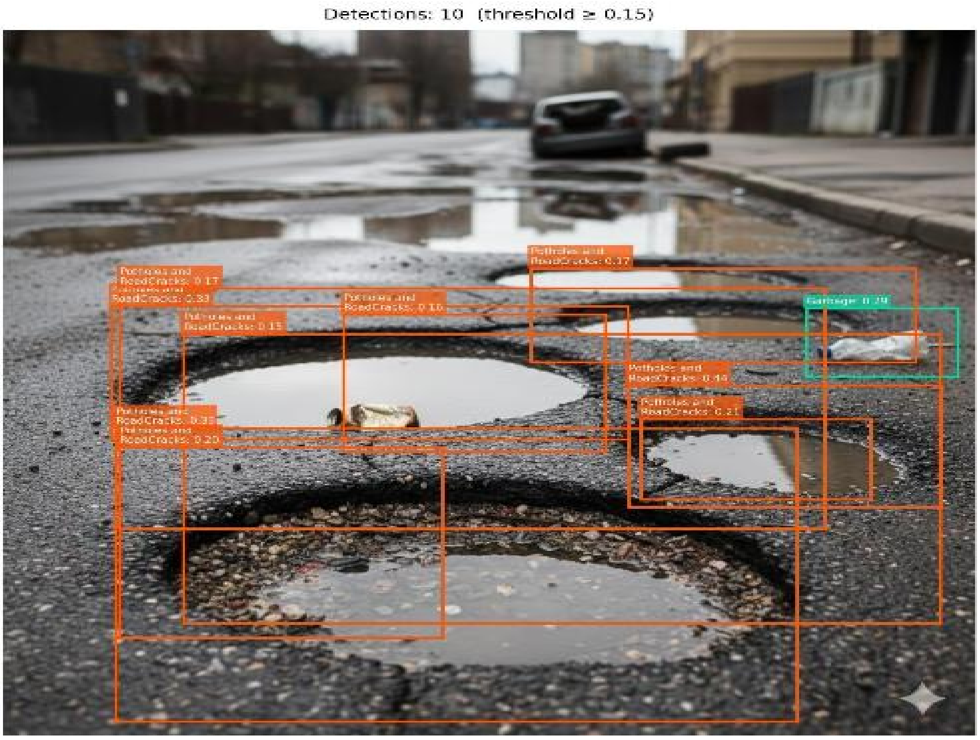
metrics/mAP50-95(B)

- RetinaNet: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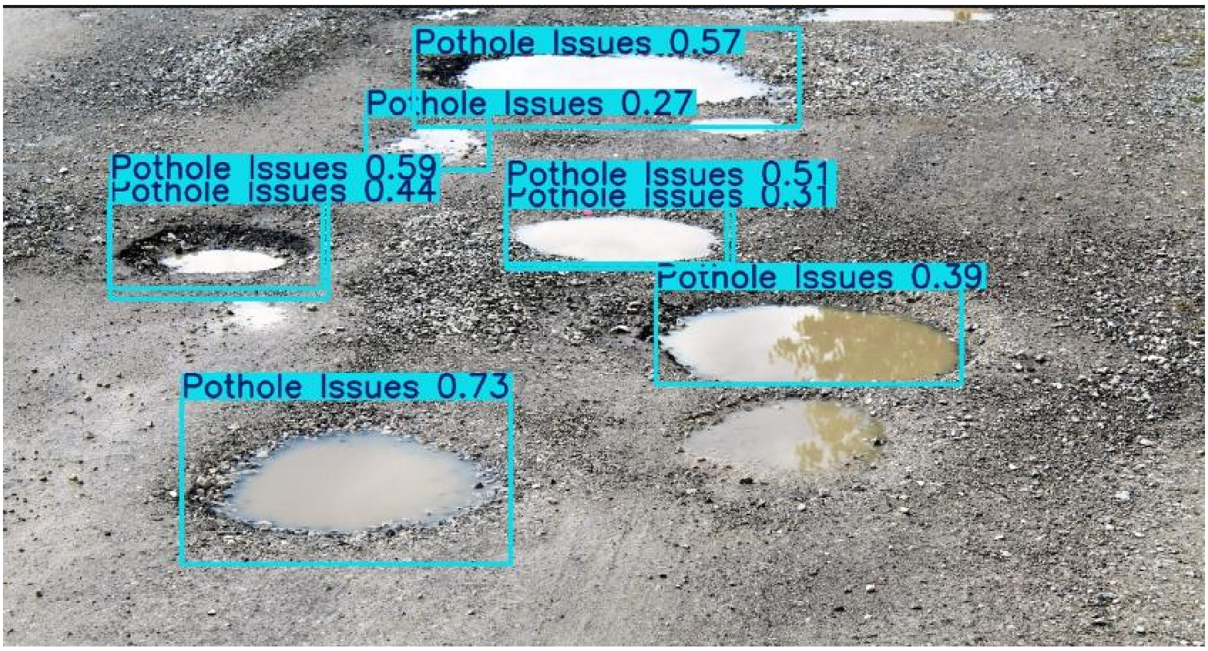
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Pothole Issues 0.51
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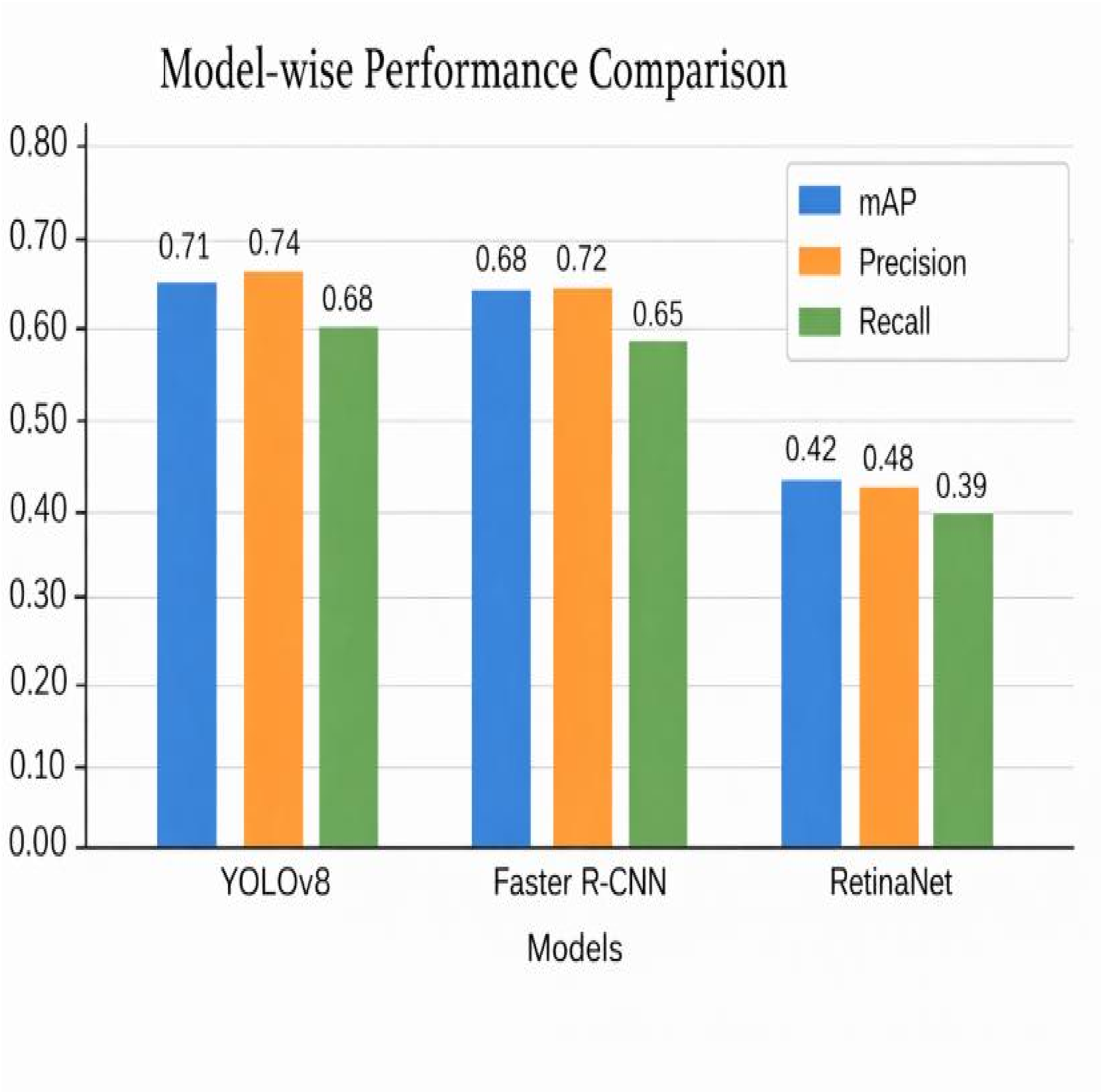
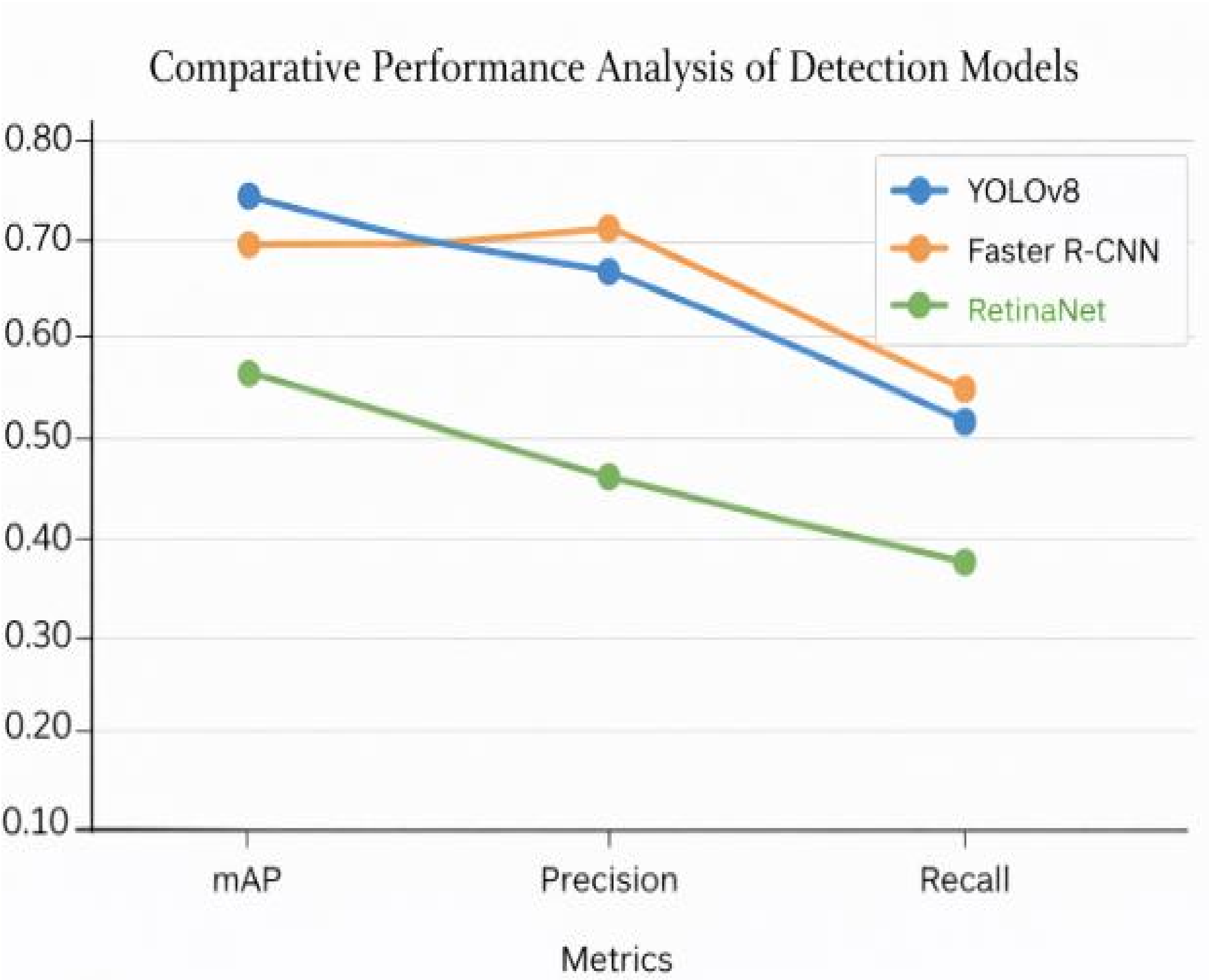
Pothole Issues 0.39
Pothole Issues 0.73

Comparison between results



STATISTICAL GRAPH



Chapter 5: Standards Adopted

Standards Adopted

This chapter outlines the design, coding, and testing standards followed during the development of the SNZI framework to ensure quality, consistency, and reproducibility.

5.1 Design Standards

The following design standards and best practices were adopted:

- IEEE Standard 1016-2009 (Software Design Descriptions) was referenced for structuring system architecture documentation.
- ISO/IEC 25010 (Systems and Software Quality Model) was used as a reference for defining quality attributes such as performance efficiency, reliability, and usability.
- UML (Unified Modeling Language) diagrams were used to represent system architecture, data flow, and module interactions.
- The dataset annotation process followed the COCO (Common Objects in Context) annotation standard for bounding box format: [x_min, y_min, width, height].
- Model evaluation followed the COCO detection metric standard, using IoU threshold of 0.50 for mAP computation (mAP@0.5).

5.2 Coding Standards

The following coding standards were observed throughout the development process:

- PEP 8 (Python Enhancement Proposal 8) style guidelines were followed for all Python code, including naming conventions, indentation (4 spaces), and line length limits (≤ 79 characters).
- Descriptive variable and function names were used to improve code readability (e.g., compute_map(), load_dataset(), apply_nms()).
- Functions were kept modular — each function was designed to perform a single, well-defined task.
- Inline comments and docstrings were provided for all major functions and classes.
- Configuration parameters (learning rate, batch size, paths) were stored in separate config files rather than hardcoded.

- Version control was maintained using Git with meaningful commit messages following the convention: [type]: description (e.g., feat: add RetinaNet training script).

5.3 Testing Standards

Testing and verification followed these standards:

- ISO/IEC 29119 (Software Testing Standard) was referenced for test case design and documentation.
- Unit testing was performed on individual preprocessing and postprocessing functions using the Python unittest framework.
- Model evaluation followed a strict train/validation/test split protocol with no data leakage between subsets.
- All experiments were logged using structured logs including timestamps, hyperparameters, and metric values to ensure reproducibility.
- Performance benchmarking was conducted on consistent hardware to ensure fair comparison across models.

Chapter 6:Conclusion and Future Scope

Conclusion and Future Scope

6.1 Conclusion

The Societal Neglected Zone Identifier (SNZI) project successfully demonstrates the viability of automated, multi-class urban infrastructure monitoring using deep learning-based object detection. By designing and evaluating a unified framework across three distinct architectures — YOLOv8, Faster R-CNN, and RetinaNet — the project provides a comprehensive comparative study that bridges the gap between academic research and practical smart city applications.

Experimental results confirm that YOLOv8 offers the best balance between detection accuracy and inference speed, achieving a mAP@0.5 of 0.71 at 12 ms per image, making it the most suitable candidate for real-time deployment. Faster R-CNN, while slightly lower in overall accuracy (mAP 0.68), provides more robust localization in complex scenes due to its two-stage proposal mechanism. RetinaNet, though underperforming in the current implementation, holds significant potential with further tuning and larger training data.

The SNZI framework lays a strong foundation for the development of intelligent urban maintenance systems that can assist municipal authorities in identifying, prioritizing, and addressing civic infrastructure issues — ultimately contributing to safer, more efficiently managed cities.

6.2 Future Scope

Several directions for future enhancement of the SNZI framework are proposed:

- Dataset Expansion: Training on a larger and more geographically diverse dataset to improve generalization across different urban environments and weather conditions.
- Severity Scoring: Introducing a severity classification module that ranks detected issues by urgency (e.g., critical pothole vs. minor crack), enabling smarter prioritization for maintenance crews.

- Real-Time Deployment: Integrating the YOLOv8 model into a mobile application or edge device (e.g., Raspberry Pi, NVIDIA Jetson Nano) for real-time on-field detection.
- Geolocation Integration: Combining detection output with GPS coordinates to automatically generate geo-tagged reports and heatmaps of neglected zones on city maps.
- Citizen Reporting Platform: Developing a web or mobile interface that allows citizens to upload images, which are then automatically analyzed by the SNZI model and reported to the relevant authorities.
- Temporal Analysis: Incorporating video stream processing to track the progression of infrastructure degradation over time.

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SOCIETAL NEGLECTED ZONE IDENTIFIER

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Abstract: The Societal Neglected Zone Identifier (SNZI) is a multi-model object detection framework designed to automatically identify urban infrastructure degradation including potholes, waterlogging, illegal dumping, vandalism, and damaged civic infrastructure. The system evaluates YOLOv8, RetinaNet, and Faster R-CNN architectures on a curated dataset of approximately 40,000 annotated images across ten neglect categories to support scalable smart city monitoring.

Individual Contribution and Findings: Mayank was responsible for model evaluation and performance benchmarking. He developed the evaluation pipeline that computed mAP, precision, recall, F1-score, and inference latency for all three models under consistent conditions. He also created visualization scripts to generate precision-recall curves and detection output visualizations. His key finding was the significant trade-off between speed and accuracy across architectures, which reinforced the case for deploying YOLOv8 in real-time urban monitoring scenarios.

Individual Contribution to Project Report Preparation: Mayank contributed to Chapter 4 (Section 4.3 — Result Analysis, Tables, and Figures) and Chapter 5 (Standards Adopted).

Individual Contribution for Project Presentation and Demonstration: Mayank presented the performance benchmarking results, explained the evaluation metrics to the panel, and demonstrated the visualization outputs during the project demonstration.

Full Signature of Supervisor: Full Signature of Student:

……………………………….. ………………………………

SOCIETAL NEGLECTED ZONE IDENTIFIER

RAJARSHI KAR

23053231

Abstract: The Societal Neglected Zone Identifier (SNZI) is a multi-model object detection framework designed to automatically identify urban infrastructure degradation including potholes, waterlogging, illegal dumping, vandalism, and damaged civic infrastructure. The system evaluates YOLOv8, RetinaNet, and Faster R-CNN architectures on a curated dataset of approximately 40,000 annotated images across ten neglect categories to support scalable smart city monitoring.

Individual Contribution and Findings: As Team Lead, Rajarshi was responsible for overall project coordination, task delegation, and timeline management. He led the architectural design of the SNZI pipeline and supervised the integration of all three detection models. He personally implemented the YOLOv8 training module, configured the training environment, and performed hyperparameter tuning. His key finding was that YOLOv8's anchor-free detection head provided a significant advantage in detecting small objects such as potholes and litter accumulation at varying image scales.

Individual Contribution to Project Report Preparation: Rajarshi contributed to Chapter 1 (Introduction), Chapter 3 (Problem Statement and System Design), and coordinated the compilation and formatting of the complete project report.

Individual Contribution for Project Presentation and Demonstration: Rajarshi led the project presentation, delivered the project overview and system architecture walkthrough, and coordinated the live demonstration of the YOLOv8 inference module.

Full Signature of Supervisor: Full Signature of Student:

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SOCIETAL NEGLECTED ZONE IDENTIFIER

HARSH SRIVASTAVA

23052322

Abstract: The Societal Neglected Zone Identifier (SNZI) is a multi-model object detection framework designed to automatically identify urban infrastructure degradation including potholes, waterlogging, illegal dumping, vandalism, and damaged civic infrastructure. The system evaluates YOLOv8, RetinaNet, and Faster R-CNN architectures on a curated dataset of approximately 40,000 annotated images across ten neglect categories to support scalable smart city monitoring.

Individual Contribution and Findings: Harsh was responsible for implementing and training the RetinaNet model. He explored focal loss configurations, backbone variants, and training schedules to optimize RetinaNet's performance on the SNZI dataset. His key finding was that RetinaNet's performance was sensitive to class imbalance in the dataset and required careful tuning of the focal loss gamma parameter. He also identified that a larger training set and extended training duration would be necessary to unlock RetinaNet's full potential on this task.

Individual Contribution to Project Report Preparation: Harsh contributed to Chapter 2 (Literature Review — Sections 2.3 and 2.5) and Chapter 4 (Section 4.2 Testing Plan and RetinaNet results).

Individual Contribution for Project Presentation and Demonstration: Harsh presented the RetinaNet implementation details and demonstrated the comparative performance analysis between RetinaNet and the other models during the evaluation.

Full Signature of Supervisor: Full Signature of Student:

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SOCIETAL NEGLECTED ZONE IDENTIFIER

PRATEEK CHANDA

23052900

Abstract: The Societal Neglected Zone Identifier (SNZI) is a multi-model object detection framework designed to automatically identify urban infrastructure degradation including potholes, waterlogging, illegal dumping, vandalism, and damaged civic infrastructure. The system evaluates YOLOv8, RetinaNet, and Faster R-CNN architectures on a curated dataset of approximately 40,000 annotated images across ten neglect categories to support scalable smart city monitoring.

Individual Contribution and Findings: Prateek was responsible for quality assurance, documentation, and future scope analysis. He reviewed the codebase for adherence to PEP 8 coding standards, managed the Git repository, and ensured proper documentation of all modules. He also researched potential deployment strategies including edge deployment on NVIDIA Jetson devices and integration with geolocation services for real-world use. His key finding was that combining SNZI output with GPS-based geo-tagging could dramatically enhance its practical utility for municipal authorities.

Individual Contribution to Project Report Preparation: Prateek contributed to Chapter 5 (Standards Adopted — Coding and Testing Standards), Chapter 6 (Conclusion and Future Scope), and was responsible for the final formatting and proofreading of the entire project report.

Individual Contribution for Project Presentation and Demonstration: Prateek contributed to the presentation slide design, presented the future scope and applications section, and assisted in demonstrating the geo-tagging and reporting extension prototype.

Full Signature of Supervisor: ….…………………………

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SOCIETAL NEGLECTED ZONE IDENTIFIER

HARSHIT JAIN

2305783

Abstract: The Societal Neglected Zone Identifier (SNZI) is a multi-model object detection framework designed to automatically identify urban infrastructure degradation including potholes, waterlogging, illegal dumping, vandalism, and damaged civic infrastructure. The system evaluates YOLOv8, RetinaNet, and Faster R-CNN architectures on a curated dataset of approximately 40,000 annotated images across ten neglect categories to support scalable smart city monitoring.

Individual Contribution and Findings: Harshit was responsible for dataset curation and preprocessing. He led the collection and organization of the ~40,000 image dataset across ten urban neglect categories. He developed the annotation conversion scripts to transform YOLO-format labels into COCO/Pascal VOC bounding box format compatible with Detectron2. His key finding was that annotation quality and label consistency significantly impacted model convergence, particularly for visually similar categories such as litter accumulation and illegal dumping.

Individual Contribution to Project Report Preparation: Harshit contributed to Chapter 2 (Literature Review — Sections 2.1, 2.2) and Chapter 3 (Section 3.1 — Project Planning and Dataset Description).

Individual Contribution for Project Presentation and Demonstration: Harshit presented the dataset construction methodology and demonstrated the annotation pipeline and preprocessing workflow during the project demonstration.

Full Signature of Supervisor: Full Signature of Student:

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SOCIETAL NEGLECTED ZONE IDENTIFIER

NIKHIL KUMAR JHA

2305716

Abstract: The Societal Neglected Zone Identifier (SNZI) is a multi-model object detection framework designed to automatically identify urban infrastructure degradation including potholes, waterlogging, illegal dumping, vandalism, and damaged civic infrastructure. The system evaluates YOLOv8, RetinaNet, and Faster R-CNN architectures on a curated dataset of approximately 40,000 annotated images across ten neglect categories to support scalable smart city monitoring.

Individual Contribution and Findings: Mayank was responsible for the implementation and training of the Faster R-CNN model using the Detectron2 framework. He configured the ResNet50 + FPN backbone, set up the Region Proposal Network, and fine-tuned training parameters including anchor scales and NMS thresholds. His key finding was that Faster R-CNN achieves strong localization accuracy on high-resolution images but suffers from significantly higher inference latency compared to single-stage models, limiting its suitability for real-time applications.

Individual Contribution to Project Report Preparation: Mayank contributed to Chapter 4 (Implementation — Sections 4.1 Faster R-CNN methodology and 4.3 Result Analysis for Faster R-CNN performance).

Individual Contribution for Project Presentation and Demonstration: Mayank demonstrated the Faster R-CNN inference results during the project presentation and explained the model's architecture and training configuration to the evaluation panel.

Full Signature of Supervisor: ....

Full Signature of Student: ....